Nitin Rathore

Bachelor of Technology
in Electrical Engineering
IIT ROPAR

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Education

Degree	Institute/Board	CGPA/Percentage	Year
Bachelor of Technology	IIT ROPAR	7.6	2019-2023
Senior Secondary	Central Board of Secondary Education	82.5%	2018
Secondary	Central Board of Secondary Education	95%	2016

Experience

- **Exl Services Pvt. Ltd.** *Oct 2023 – Present*
  Data Scientist

  –**Agentic AI Framework:** Built modular autonomous workflows using LangChain/LangGraph enabling multi-step reasoning, memory, tool-use and decision automation.

  –**RAG Systems:** Designed scalable RAG pipelines integrating dense retrieval + LLMs producing high-accuracy, context-aware responses.resulted in faster inference and improved accuracy.

  –**AI Summarizer:** Developed an LLM-powered granular summarization system generating structured multi-section insights including key points, sentiment and contextual highlights.

  –**AI Evaluator (Agent Capability Scoring):** Built an automated evaluator that scores AI agent reasoning, task efficiency, tool usage and factual correctness using LLM-based rubrics.

  –**Google Conversational Insights Pipeline:** Developed an end-to-end conversation analytics pipeline delivering insights such as topics, summaries, sentiment, AI scorecards, and performance metrics at scale.

  –**Fine-Tuning Transformer Models:**

  ∗**Summarization (BART):** Fine-tuned a BART model on custom datasets for abstractive text summarization, and evaluated performance using ROUGE metrics to measure overlap with reference summaries.

  ∗**Emotion Classification (RoBERTa Base):** Fine-tuned RoBERTa Base on an emotion-labeled dataset, achieving an average recall score of **93%**, indicating strong performance in identifying emotional categories.

  –**Unsupervised Learning:** Implemented topic modeling using BERTopic, leveraging techniques such as UMAP for dimensionality reduction, HDBSCAN for clustering, and KeyBERT for keyword extraction. Enhanced topic representation and interpretability through representation modeling and LLM-based topic naming, enabling meaningful insights from unstructured text data achieving an accuracy of 85% through manual validation.

  –**Named Entity Recognition:** Fine-tuned BERT-based NER models with custom datasets; implemented reverse QA for enriched entity extraction achieving an accuracy of more than 90% in detection of the desired entities.

  –**LLM Engineering:** Created robust LLM-based solutions using CoT, ReAct, and structured prompting.Designed Prompt-enginerring strategies that transformed LLM behaviour from majority class bias to balanced prediction,enabling reliable churn detection of all different classes with 87% recall post - Finetuning prompt.

Projects

- **RAG + LLM based Document Q&A System** *Jan 2024 – Present*
  Streamlit, LangChain **Deployed Project Link**
  – Developed and deployed an end-to-end RAG (Retrieval-Augmented Generation) powered application on Streamlit, enabling users to upload documents (PDF/TXT/DOCX) and interact with them via natural language queries.
  – Integrated LLMs with RAG for dense retrieval for contextually relevant responses.
  – Designed a clean, user-friendly Streamlit UI to manage document upload, embedding generation, and real-time query answering.
  – Hosted the interactive application publicly, accessible at the link above.

Technical Skills

- **Programming Languages**: Jupyter Notebook, Python,langchain,langgraph
- **data science development**: LLM,pytorch,neural network,LM,generative AI,Sagemaker,Bedrock,GCP,Vertex Ai

Key courses taken

- **Data science**: Statics, Power BI, Machine learning, Neural network.
- **CSE & Maths**: Algorithm & Data Structure(geeks for geeks), Object Oriented Programming,Web developer course(coding ninja).